

Si53340-45 Data Sheet

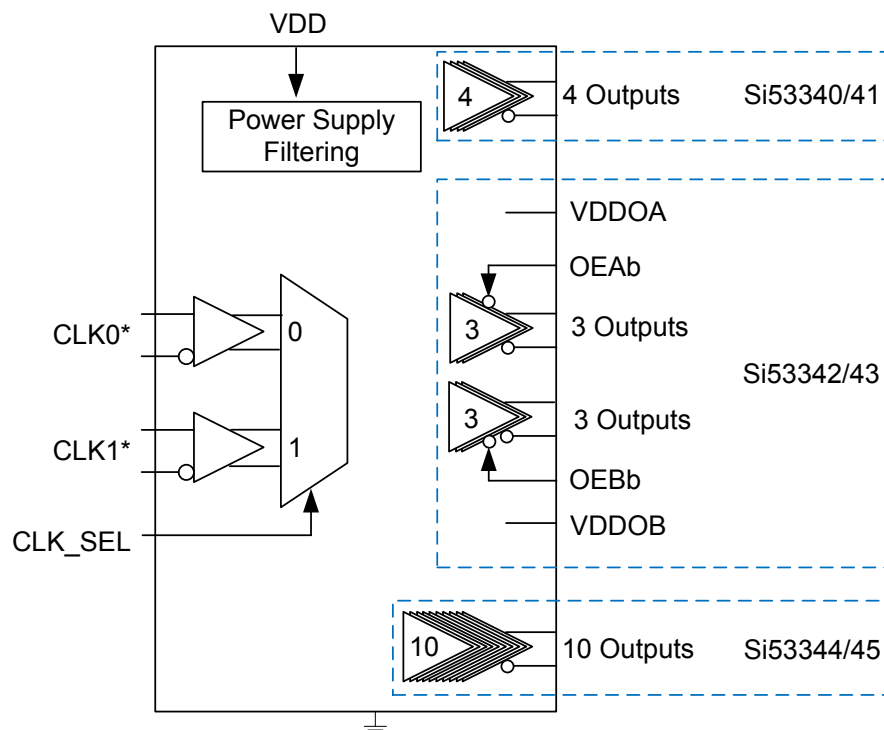
Low-Jitter LVDS Fanout Clock **Buffers** with up to 10 LVDS Outputs from Any-Format Input and Wide Frequency Range from dc up to 1250 MHz

The Si53340-45 family of LVDS fanout buffers is ideal for clock/data distribution and redundant clocking applications. These devices feature typical ultra-low jitter of 50 fs and operate over a wide frequency range from dc to 1250 MHz. Built-in LDOs deliver high PSRR performance and reduces the need for external components simplifying low jitter clock distribution in noisy environments.

They are available in multiple configurations and offer a selectable input clock using a 2:1 input mux. Other features include independent output enable and built-in format translation. These buffers can be paired with the Si534x clocks and Si5xx oscillators to deliver end-to-end clock tree performance.

KEY FEATURES

- Ultra-low additive jitter: 50 fs rms
- Built-in LDOs for high PSRR performance
- Up to 10 LVDS Outputs
- Any-format Inputs (LVPECL, Low-Power LVPECL, LVDS, CML, HCSL, LVCMOS)
- Wide frequency range: dc to 1250 MHz
- Output Enable option
- Multiple configuration options
- 2:1 Input Mux
- RoHS compliant, Pb-free
- **Temperature range: -40 to +85 °C**



*Si53341/43/45 require Single-ended Inputs

1. Ordering Guide

Table 1.1. SI5334x Ordering Guide

Part Number	Input	LVDS Output	Output Enable (OE)	Frequency Range	Package
SI53340-B-GM	2:1 selectable MUX Any-format	1 bank / 4 Outputs	—	dc to 1250 MHz	16-QFN 3 x 3 mm
SI53341-B-GM	2:1 selectable MUX LVCMOS	1 bank / 4 Outputs	—	dc to 200 MHz	16-QFN 3 x 3 mm
SI53342-B-GM	2:1 selectable MUX Any-format	2 banks / 3 Outputs	1 per bank	dc to 1250 MHz	24-QFN 4 x 4 mm
SI53343-B-GM	2:1 selectable MUX LVCMOS	2 banks / 3 Outputs	1 per bank	dc to 200 MHz	24-QFN 4 x 4 mm
SI53344-B-GM	2:1 selectable MUX Any-format	1 bank / 10 Outputs	—	dc to 1250 MHz	32-QFN 5 x 5 mm
SI53345-B-GM	2:1 selectable MUX LVCMOS	1 bank / 10 Outputs	—	dc to 200 MHz	32-QFN 5 x 5 mm

2. Functional Description

The Si53340-45 are a family of low-jitter, low skew, fixed format (LVDS) buffers. The Si53340/42/44 have a universal input that accepts most common differential or LVCMOS input signals, while the Si53341/43/45 accept only LVCMOS inputs. These devices are available in multiple configurations customized for the end application (refer to [1. Ordering Guide](#) for more details on configurations).

2.1 Universal, Any-Format Input Termination (Si53340/42/44)

The universal input stage enables simple interfacing to a wide variety of clock formats, including LVPECL, Low-power LVPECL, LVDS, CML, HCSL, and LVCMOS. The tables below summarize the various ac- and dc-coupling options supported by the device. For the best high-speed performance, the use of differential formats is recommended. For both single-ended and differential input clocks, the fastest possible slew rate is recommended since low slew rates can increase the noise floor and degrade jitter performance. Though not required, a minimum slew rate of 0.75 V/ns is recommended for differential formats and 1.0 V/ns for single-ended formats. See “AN766: Understanding and Optimizing Clock Buffer’s Additive Jitter Performance” for more information.

Table 2.1. Clock Input Options

Clock Format	1.8 V	2.5/3.3 V
AC-Coupled		
LVPECL/Low-power LVPECL	N/A	Yes
LVCMOS	No	Yes
LVDS	Yes	Yes
HCSL	No	Yes (3.3 V)
CML	Yes	Yes
DC-Coupled		
LVPECL/Low-power LVPECL	N/A	Yes
LVCMOS	No	Yes
LVDS	No	Yes
HCSL	No	Yes (3.3 V)
CML	No	No

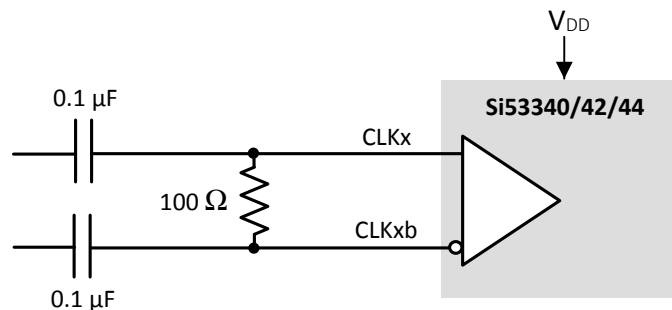


Figure 2.1. Differential (HCSL, LVPECL, Low-Power LVPECL, LVDS, CML) AC-Coupled Input Termination

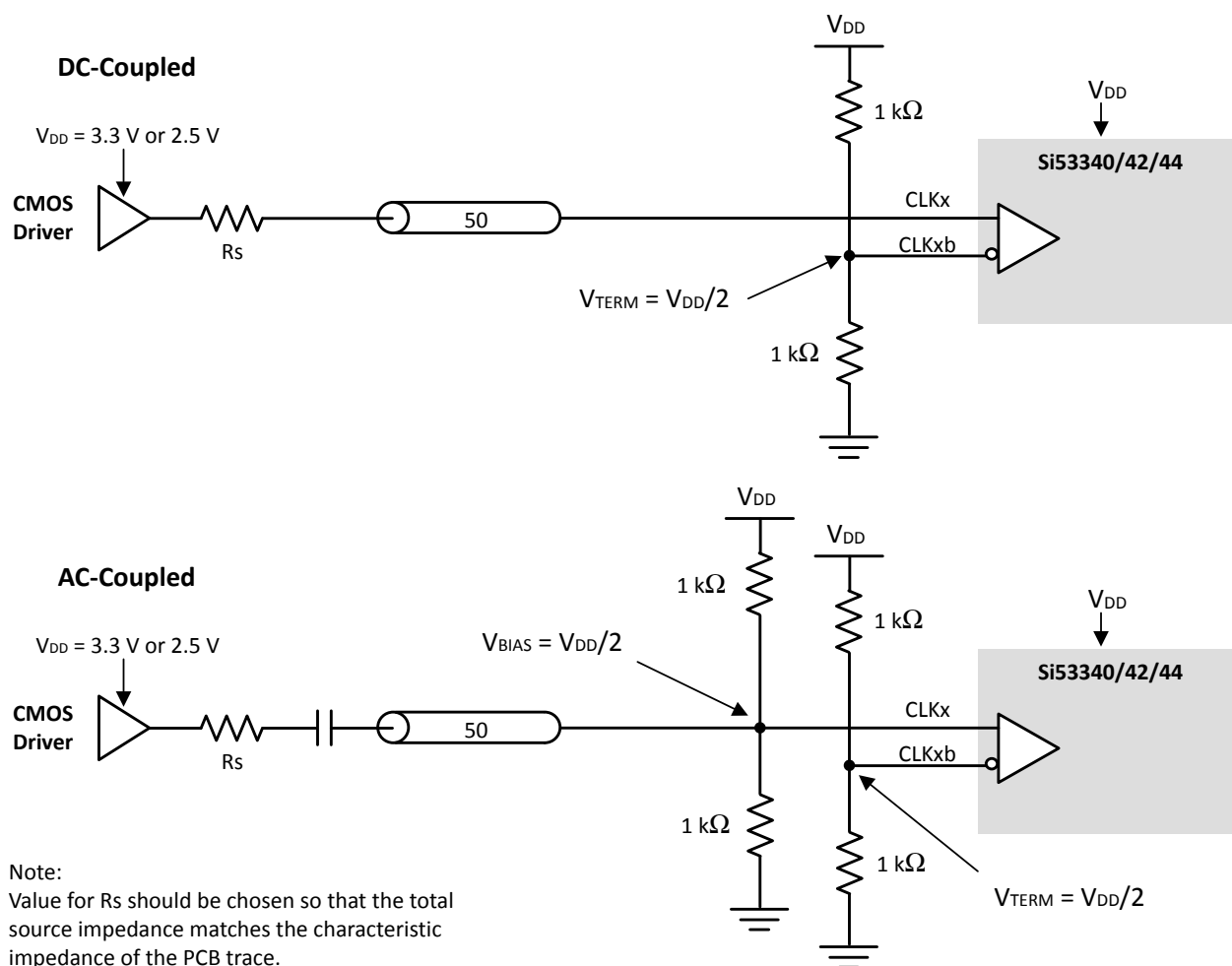
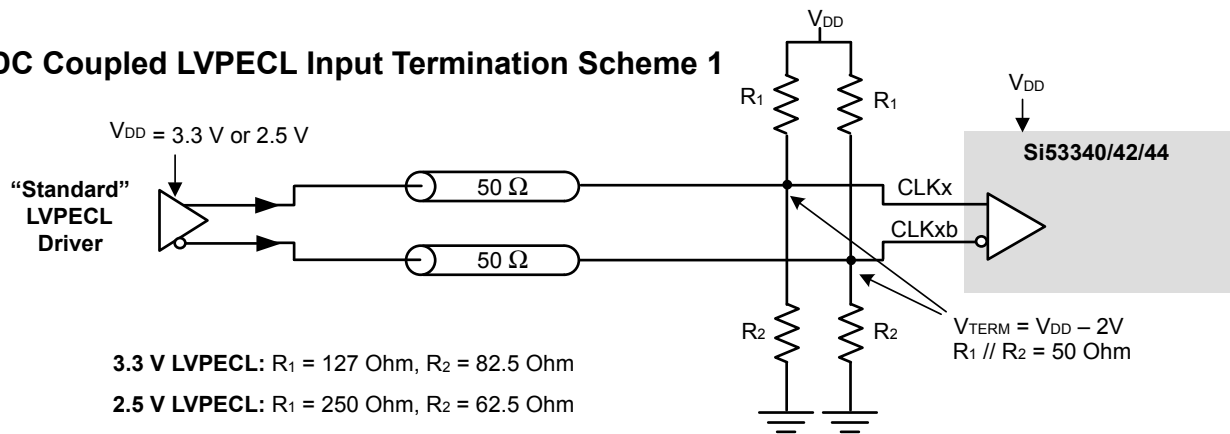
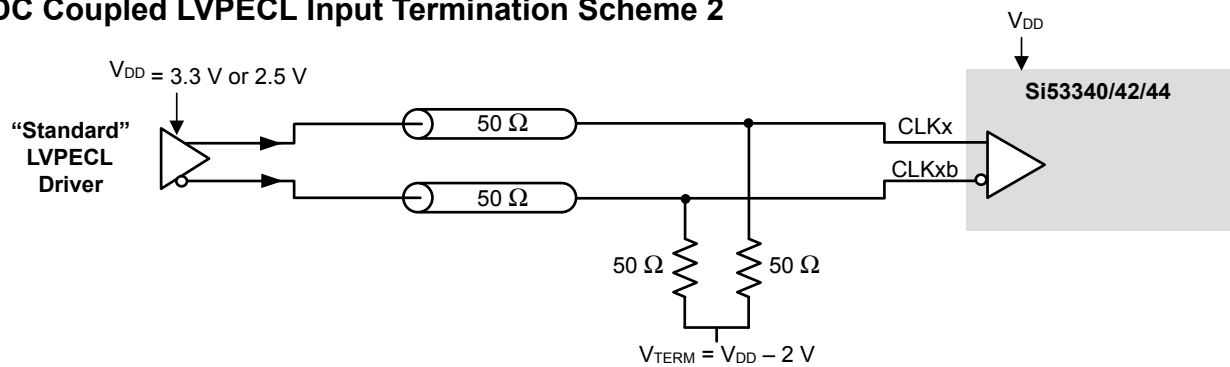


Figure 2.2. Single-Ended (LVCMOS) Input Termination

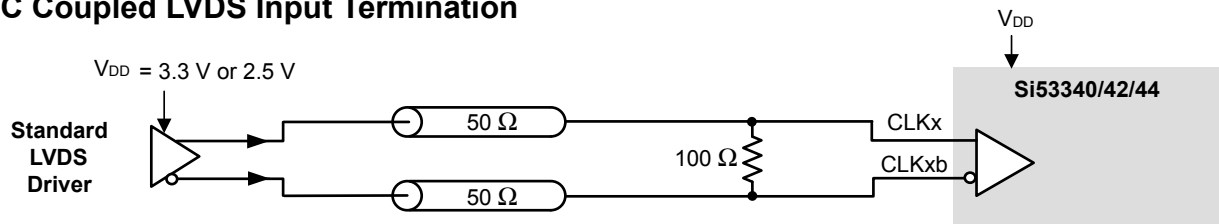
DC Coupled LVPECL Input Termination Scheme 1



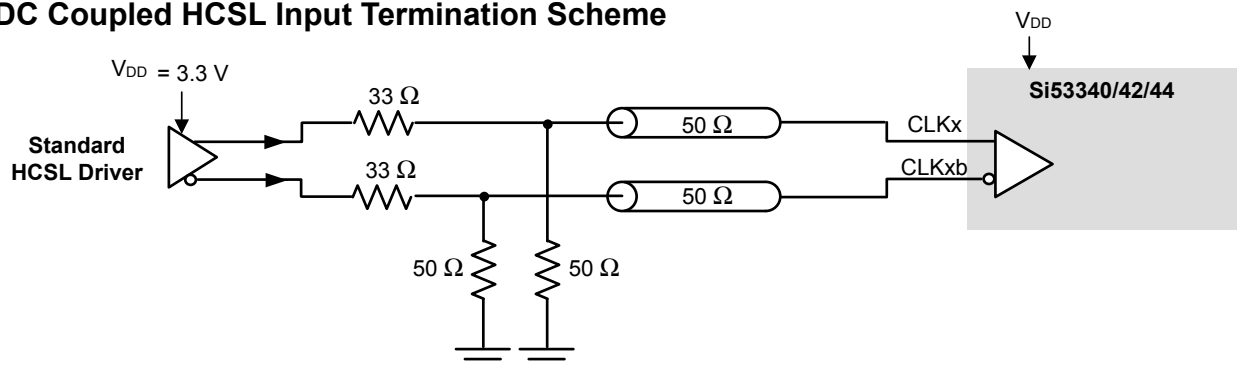
DC Coupled LVPECL Input Termination Scheme 2



DC Coupled LVDS Input Termination



DC Coupled HCSL Input Termination Scheme



Note: 33 Ohm series termination is optional depending on the location of the receiver.

Figure 2.3. Differential DC-Coupled Input Terminations

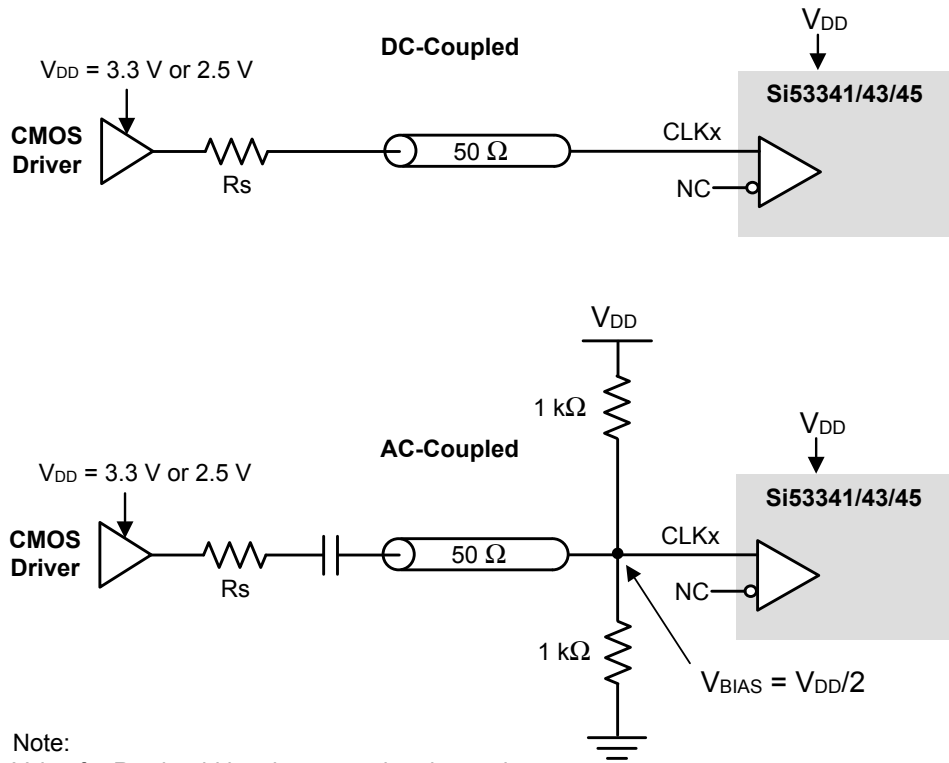
2.2 LVCMOS Input Termination (Si53341/43/45)

The table below summarizes the various ac- and dc-coupling options supported by the LVCMOS device, and the figure shows the recommended input clock termination.

Note: 1.8V LVCMOS inputs are not supported for Si53341/43/45.

Table 2.2. LVCMOS Input Clock Options

	LVCMOS	
	AC-Coupled	DC-Coupled
1.8 V	No	No
2.5/3.3 V	Yes	Yes



Note:
Value for R_s should be chosen so that the total source impedance matches the characteristic impedance of the PCB trace.

Figure 2.4. Recommended Input Clock Termination (Si53341/43/45)

2.3 Input Bias Resistors

Internal bias resistors ensure a differential output low condition in the event that the clock inputs are not connected. The non-inverting input is biased with a 18.75 kΩ pull-down to GND and a 75 kΩ pull-up to V_{DD}. The inverting input is biased with a 75 kΩ pull-up to V_{DD}.

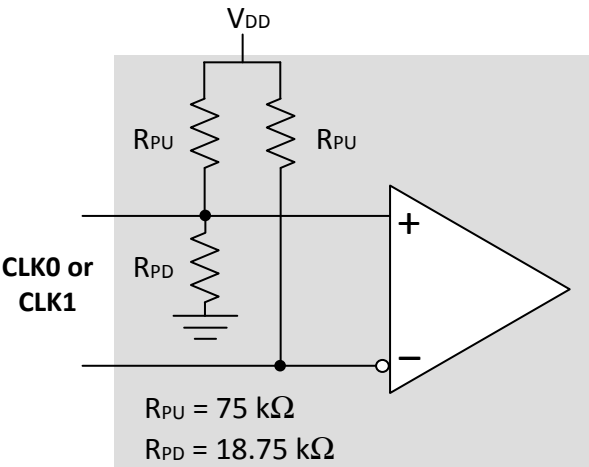


Figure 2.5. Input Bias Resistors

Note: To minimize the possibility of system noise coupling into the Si5334x differential inputs and adversely affecting the buffered output, Skyworks recommends 1 PPS clocks and disabled/gapped clocks be DC-coupled and driven “stop-low” .

2.4 Input Mux

The Si5334x provide two clock inputs for applications that need to select between one of two clock sources. The CLK_SEL pin selects the active clock input. The following table summarizes the input and output clock based on the input mux and output enable pin settings.

Table 2.3. Input Mux Logic

CLK_SEL	CLK0	CLK1	Q ¹	Qb
L	L	X	L	H
L	H	X	H	L
H	X	L	L	H
H	X	H	H	L

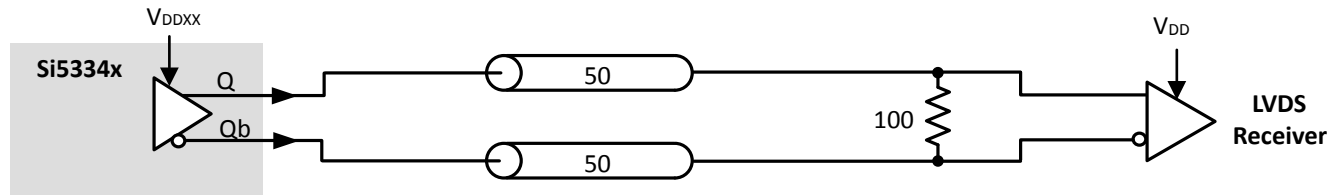
Note:

- 1. On the next negative transition of CLK0 or CLK1.

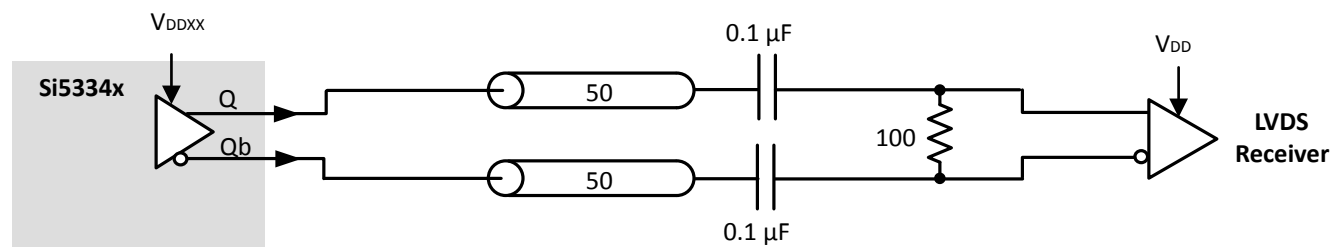
2.5 Output Clock Termination Options

The recommended output clock termination options are shown below. Unused outputs should be left unconnected.

DC Coupled LVDS Termination



AC Coupled LVDS Termination



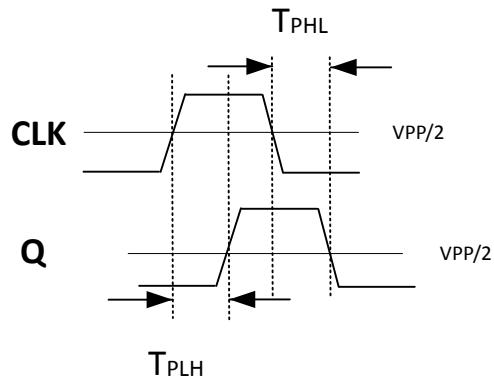
Note:

For Si53340/41/44/45, $V_{DDXX} = V_{DD} = 3.3\text{ V}, 2.5\text{ V}, 1.8\text{ V}$

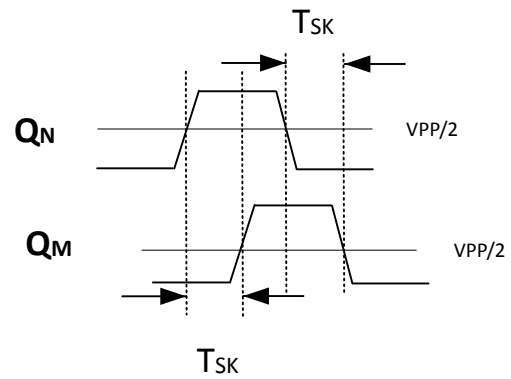
For Si53342/43, $V_{DDXX} = V_{DDOA}$ Or $V_{DDOB} = 3.3\text{ V}, 2.5\text{ V}, 1.8\text{ V}$

Figure 2.6. LVDS Output Terminations

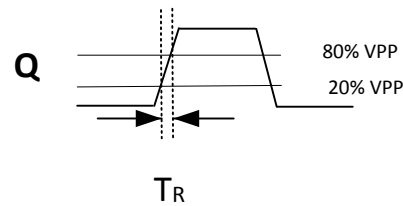
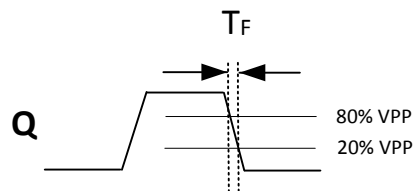
2.6 AC Timing Waveforms



Propagation Delay



Output-Output Skew



Rise/Fall Time

Figure 2.7. AC Timing Waveforms

2.7 Typical Phase Noise Performance: Differential Input Clock

Each of the three phase noise plots superimposes Source Jitter, Total SE Jitter and Total Diff Jitter on the same diagram.

- **Source Jitter**—Reference clock phase noise (measured Single-ended to PNA).
- **Total Jitter (SE)**—Combined source and clock buffer phase noise measured as a single-ended output to the phase noise analyzer and integrated from 12 kHz to 20 MHz.
- **Total Jitter (Diff)**—Combined source and clock buffer phase noise measured as a differential output to the phase noise analyzer and integrated from 12 kHz to 20 MHz. The differential measurement as shown in each figure is made using a balun. For more information, see 3. [Electrical Specifications](#).

Note: To calculate the total RMS phase jitter when adding a buffer to your clock tree, use the root-sum-square (RSS).

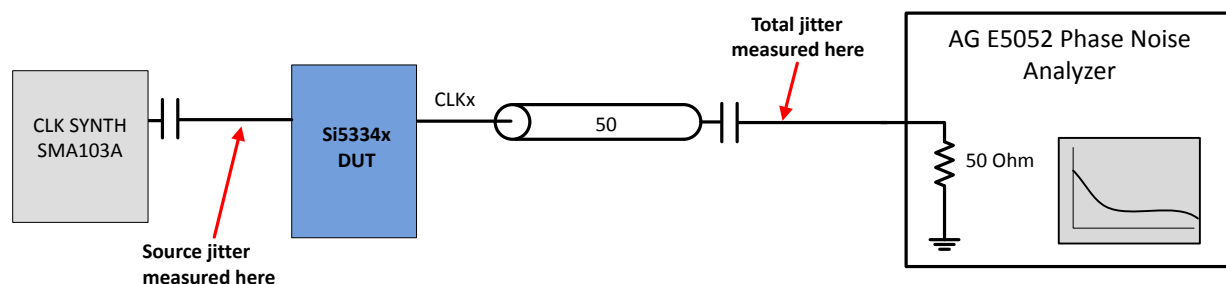
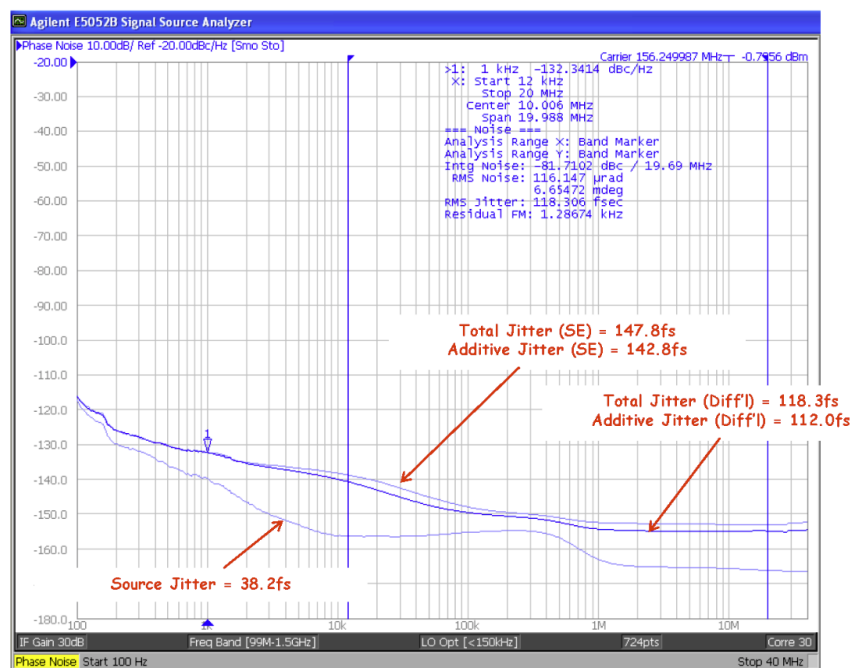


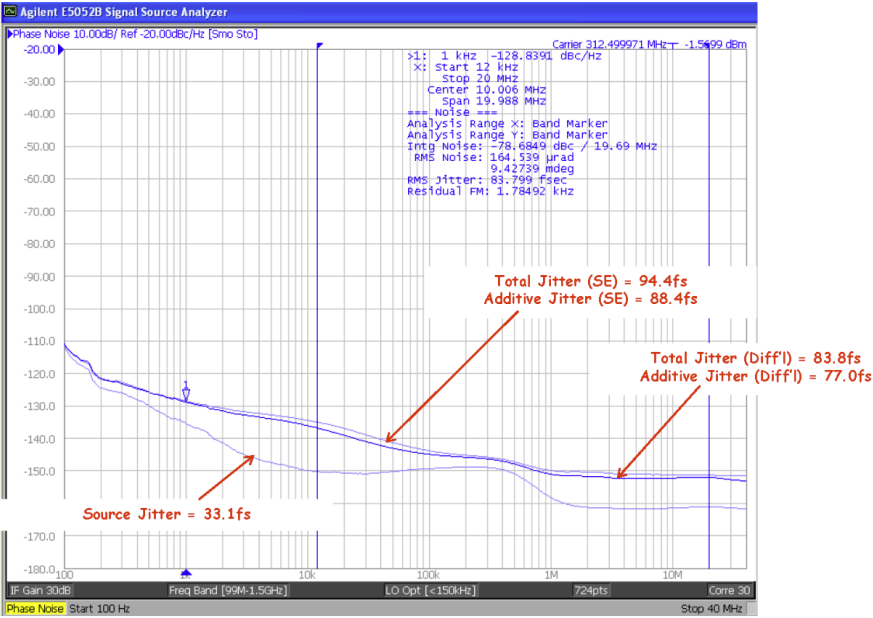
Figure 2.8. Differential Measurement Method Using a Balun

The total jitter is a measure of the source plus the buffer's additive phase jitter. The additive jitter (rms) of the buffer can then be calculated (via root-sum-square addition).



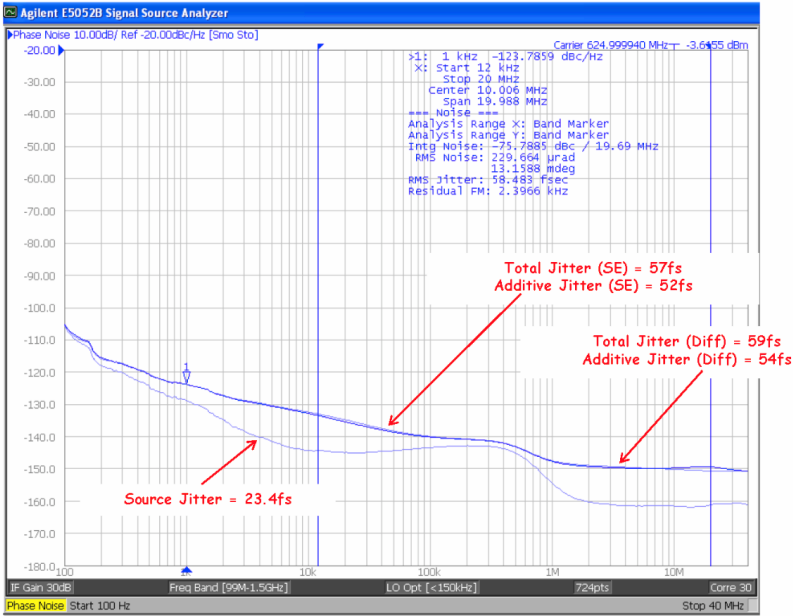
Frequency (MHz)	Differential Input Slew Rate (V/ns)	Source Jitter (fs)	Total Jitter (SE) (fs)	Additive Jitter (SE) (fs)	Total Jitter (Differential) (fs)	Additive Jitter (Differential) (fs)
156.25	1.0	38.2	147.8	142.8	118.3	112.0

Figure 2.9. Total Jitter Differential Input (156.25 MHz)



Frequency (MHz)	Differential Input Slew Rate (V/ns)	Source Jitter (fs)	Total Jitter (SE) (fs)	Additive Jitter (SE) (fs)	Total Jitter (Differential) (fs)	Additive Jitter (Differential) (fs)
312.5	1.0	33.10	94.39	88.39	83.80	76.99

Figure 2.10. Total Jitter Differential Input (312.5 MHz)



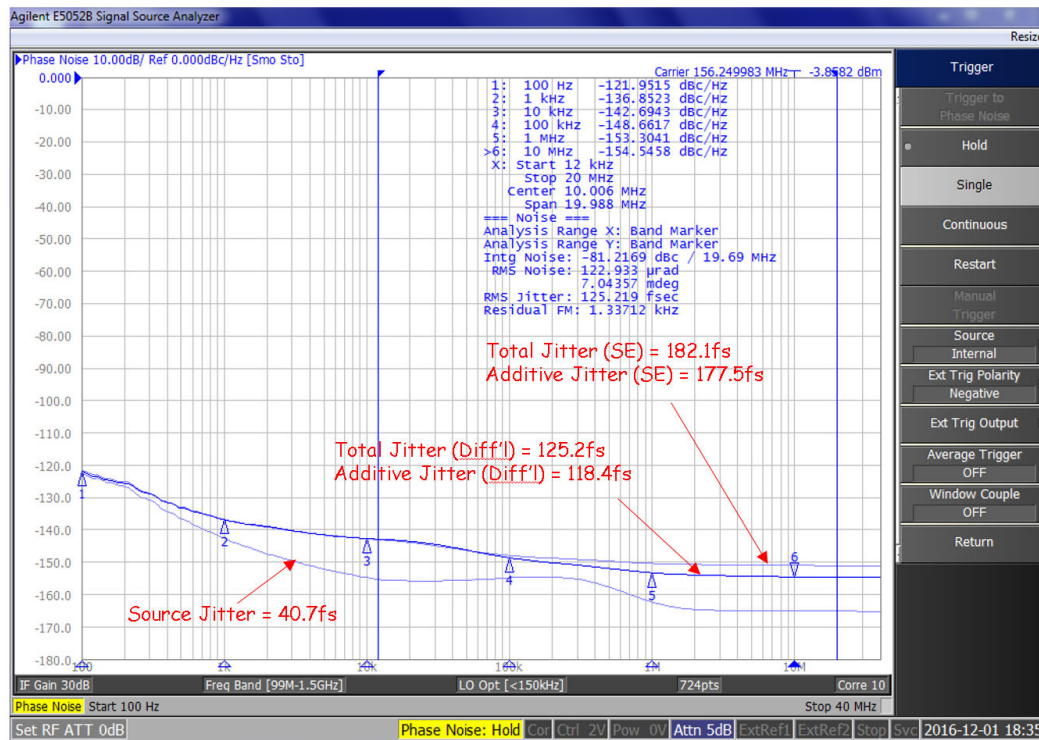
Frequency (MHz)	Differential Input Slew Rate (V/ns)	Source Jitter (fs)	Total Jitter (SE) (fs)	Additive Jitter (SE) (fs)	Total Jitter (Differential) (fs)	Additive Jitter (Differential) (fs)
625	1.0	23	57	52	59	54

Figure 2.11. Total Jitter Differential Input (625 MHz)

2.8 Typical Phase Noise Performance: Single-Ended Input Clock

For single-ended input phase noise measurements, the input was connected directly without the use of a balun.

The following figure shows three phase noise plots superimposed on the same diagram.



Frequency (MHz)	Single-Ended Input Slew Rate (V/ns)	Source Jitter (fs)	Total Jitter (SE) (fs)	Additive Jitter (SE) (fs)	Total Jitter (Differential) (fs)	Additive Jitter (Differential) (fs)
156.25	1.0	40.74	182.12	177.51	125.22	118.41

Figure 2.12. Total Jitter Single-Ended Input (156.25 MHz)

2.9 Input Mux Noise Isolation

The input clock mux is designed to minimize crosstalk between the CLK0 and CLK1. This improves phase jitter performance when clocks are present at both the CLK0 and CLK1 inputs. The following figure shows a measurement of the input mux's noise isolation.

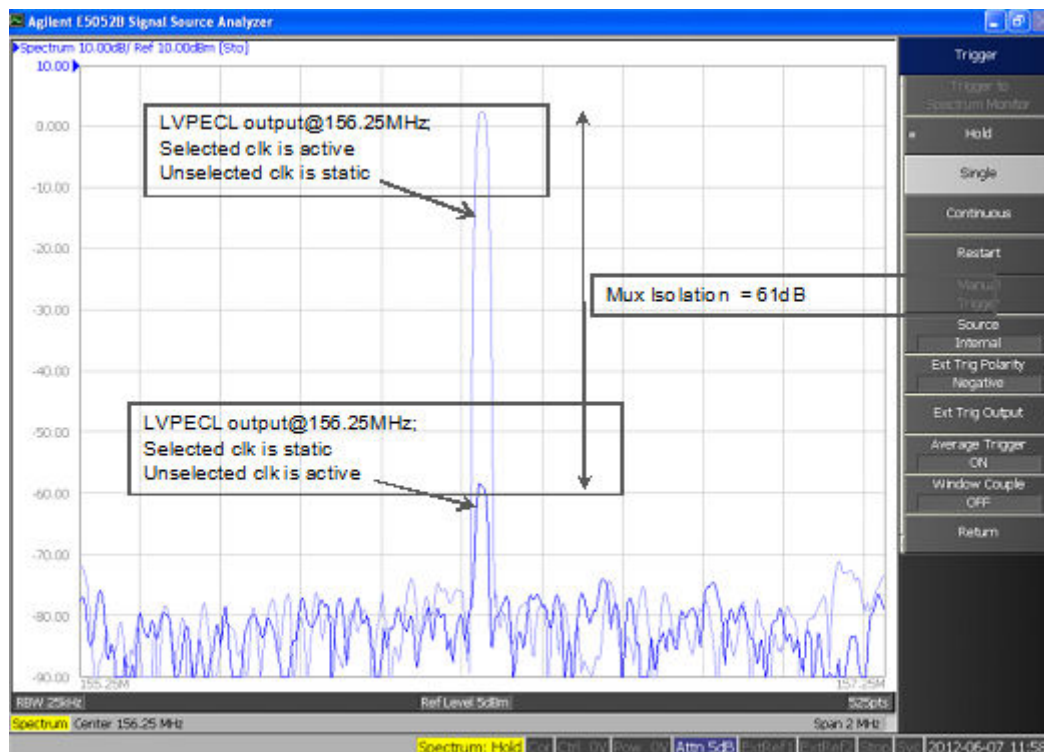


Figure 2.13. Input Mux Noise Isolation (Differential Input Clock, 44-QFN Package)

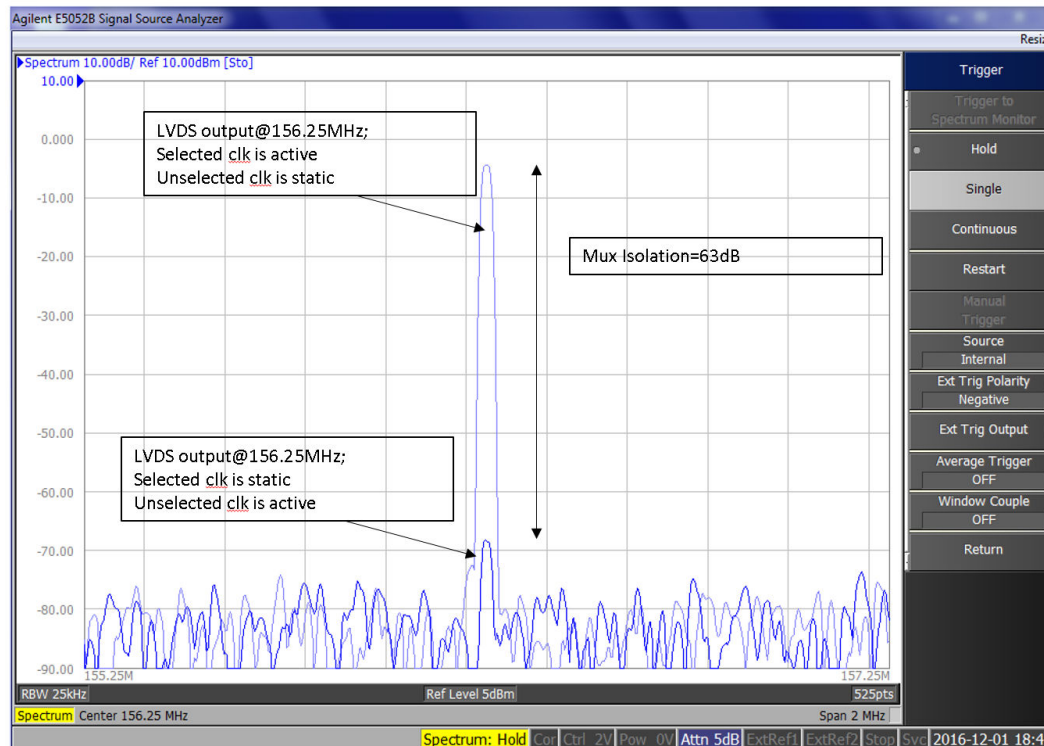


Figure 2.14. Input Mux Noise Isolation (Single-Ended Input Clock, 24-QFN Package)

2.10 Power Supply Noise Rejection

The device supports on-chip supply voltage regulation to reject power supply noise and simplify low-jitter operation in real-world environments. This feature enables robust operation alongside FPGAs, ASICs and SoCs and may reduce board-level filtering requirements. See “[AN491: Power Supply Rejection for Low-Jitter Clocks](#)” for more information.

3. Electrical Specifications

Table 3.1. Recommended Operating Conditions

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Ambient Operating Temperature	T_A		−40	—	85	°C
Supply Voltage Range	V_{DD}	LVDS	1.71	1.8	1.89	V
			2.38	2.5	2.63	V
			2.97	3.3	3.63	V

Table 3.2. Input Clock Specifications
 $V_{DD} = 1.8 \text{ V}, 2.5 \text{ V}, \text{ or } 3.3 \text{ V}; T_A = -40 \text{ to } 85 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Differential Input Common Mode Voltage	V_{CM}		0.05	—	—	V
Differential Input Swing (peak-to-peak)	V_{IN}		0.2	—	2.2	V
Input High Voltage	V_{IH}		$V_{DD} \times 0.7$	—	—	V
Input Low Voltage	V_{IL}		—	—	$V_{DD} \times 0.3$	V
Input Capacitance	C_{IN}	CLK0 and CLK1 pins with respect to GND	—	5	—	pF

Table 3.3. DC Common Characteristics
 $V_{DD} = 1.8 \text{ V}, 2.5 \text{ V}, \text{ or } 3.3 \text{ V}; T_A = -40 \text{ to } 85 \text{ }^{\circ}\text{C}$

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Core Supply Current	I_{DD}^1	Si53340/41	—	140	—	mA
		Si53342/43	—	80	—	mA
		Si53344/45	—	280	—	mA
Output Supply Current (Per Clock Output)	I_{DDO}^1	Si53342/43	—	21	—	mA
Input High Voltage	V_{IH}	CLK_SEL, OEAb, OEBb	$V_{DD} \times 0.8$	—	—	V
Input Low Voltage	V_{IL}	CLK_SEL, OEAb, OEBb	—	—	$V_{DD} \times 0.2$	V
Internal Pull-down Resistor	R_{DOWN}	CLK_SEL, OEAb, OEBb	—	25	—	k Ω

Note:

 1. Measured using ac-coupled termination at $V_{DD}/V_{DDOX} = 3.3 \text{ V}$.

Table 3.4. Output Characteristics (LVDS) $V_{DD} = 1.8\text{ V}$, 2.5 V , or 3.3 V ; $T_A = -40$ to $85\text{ }^{\circ}\text{C}$

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Single-Ended Output Swing ¹	V_{SE}	$R_L = 100\ \Omega$ across Q_N and Q_{bN}	200	—	490	mV
Output Common Mode Voltage ($V_{DD} = 2.5$ or 3.3 V)	V_{COM1}	$V_{DD} = 2.38$ to 2.63 V , 2.97 to 3.63 V , $R_L = 100\ \Omega$ across Q_N and Q_{bN}	1.10	1.25	1.35	V
Output Common Mode Voltage ($V_{DD} = 1.8\text{ V}$)	V_{COM2}	$V_{DD} = 1.71$ to 1.89 V , $R_L = 100\ \Omega$ across Q_N and Q_{bN}	0.83	0.97	1.25	V
Note: 1. Unused outputs can be left floating. Do not short unused outputs to ground.						

Table 3.5. AC Characteristics $V_{DD} = 1.8\text{ V} \pm 5\%$, $2.5\text{ V} \pm 5\%$, or $3.3\text{ V} \pm 10\%$; $T_A = -40$ to $85\text{ }^{\circ}\text{C}$

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Frequency	F	Si53341/43/45	dc	—	200	MHz
		Si53340/42/44	dc	—	1250	MHz
Duty Cycle (50% input duty cycle)	D_C	20/80% $T_R/T_F < 10\%$ of period Differential input clock	47	50	53	%
		20/80% $T_R/T_F < 10\%$ of period (Single-ended input clock)	45	50	55	%
Minimum Input Clock Slew Rate	SR_{diff}	Required to meet prop delay and ad- ditive jitter specifications (20–80%)	0.75	—	—	V/ns
	SR_{se}	Required to meet prop delay and ad- ditive jitter specifications (20–80%)	1.00	—	—	V/ns
Output Rise/Fall Time	T_R/T_F	20–80%	—	—	350	ps
Minimum Input Pulse Width	T_W		360	—	—	ps
Propagation Delay	T_{PLH} , T_{PHL}		650	850	1050	ns
Output-to-Output Skew ¹	T_{SK}		—	—	50	ps
Part-to-Part Skew ²	T_{PS}		—	—	125	ps
Power Supply Noise Rejection ³	PSRR	10 kHz sinusoidal noise	—	–70	—	dBc
		100 kHz sinusoidal noise	—	–65	—	dBc
		500 kHz sinusoidal noise	—	–60	—	dBc
		1 MHz sinusoidal noise	—	–57.5	—	dBc

Note:

- Output-to-output skew specified for outputs with identical configuration.
- Defined as skew between any output on different devices operating at the same supply voltage, temperature, and equal load condition. Using the same type of inputs on each device, the outputs are measured at the differential cross points.
- Measured for 156.25 MHz carrier frequency. Sine-wave noise added to V_{DD} ($3.3\text{ V} = 100\text{ mV}_{PP}$) and noise spur amplitude measured. See “AN491: Power Supply Rejection for Low-Jitter Clocks” for more information.

Table 3.6. Additive Jitter, Differential Clock Input

V _{DD}	Input ^{1, 2}				Output	Additive Jitter (fs rms, 12 kHz to 20 MHz) ³	
	Freq (MHz)	Clock Format	Amplitude V _{IN} (Single-Ended, Peak-to-Peak)	Differential 20% to 80% Slew Rate (V/ns)	Clock Format	Typ	Max
3.3	725	Differential	0.15	0.637	LVDS	50	65
3.3	156.25	Differential	0.5	0.458	LVDS	150	200
2.5	725	Differential	0.15	0.637	LVDS	50	65
2.5	156.25	Differential	0.5	0.458	LVDS	145	195

Note:

1. For best additive jitter results, use the fastest slew rate possible. See “[AN766: Understanding and Optimizing Clock Buffer's Additive Jitter Performance](#)” for more information.
2. AC-coupled differential inputs.
3. Measured differentially using a balun at the phase noise analyzer input. See Figure 1.

Table 3.7. Additive Jitter, Single-Ended Clock Input

V _{DD}	Input ^{1, 2}				Output	Additive Jitter (fs rms, 12 kHz to 20 MHz) ³	
	Freq (MHz)	Clock Format	Amplitude V _{IN} (Single-Ended, Peak-to-Peak)	Single-Ended 20% to 80% Slew Rate (V/ns)	Clock Format	Typ	Max
3.3	156.25	Single-ended	2.18	1	LVDS	150	200
2.5	156.25	Single-ended	2.18	1	LVDS	145	195

Note:

1. For best additive jitter results, use the fastest slew rate possible. See “[AN766: Understanding and Optimizing Clock Buffer's Additive Jitter Performance](#)” for more information.
2. DC-coupled single-ended inputs.
3. Measured differentially using a balun at the phase noise analyzer input. See figure below.

Table 3.8. Thermal Conditions

Parameter	Symbol	Test Condition	Value	Unit
16-QFN Thermal Resistance, Junction to Ambient	θ_{JA}	Still air	57.6	°C/W
16-QFN Thermal Resistance, Junction to Case	θ_{JC}	Still air	41.5	°C/W
24-QFN Thermal Resistance, Junction to Ambient	θ_{JA}	Still air	37	°C/W
24-QFN Thermal Resistance, Junction to Case	θ_{JC}	Still air	25	°C/W
32-QFN Thermal Resistance, Junction to Ambient	θ_{JA}	Still air	99.6	°C/W
32-QFN Thermal Resistance, Junction to Case	θ_{JC}	Still air	10.3	°C/W

Table 3.9. Absolute Maximum Ratings¹

Parameter	Symbol	Test Condition	Min	Typ	Max	Unit
Storage Temperature	T_S		–55	—	150	°C
Supply Voltage	V_{DD}		–0.5	—	3.8	V
Input Voltage	V_{IN}		–0.5	—	$V_{DD} + 0.3$	V
Output Voltage	V_{OUT}		—	—	$V_{DD} + 0.3$	V
ESD Sensitivity	HBM	HBM, 100 pF, 1.5 k Ω	—	—	2000	V
	CDM		—	—	500	V
Peak Soldering Reflow Temperature	T_{PEAK}	Pb-Free; Solder reflow profile per JEDEC J-STD-020	—	—	260	°C
Maximum Junction Temperature	T_J		—	—	125	°C

Note:

1. Stresses beyond those listed in this table may cause permanent damage to the device. Functional operation specification compliance is not implied at these conditions. Exposure to maximum rating conditions for extended periods may affect device reliability.

4. Detailed Block Diagrams

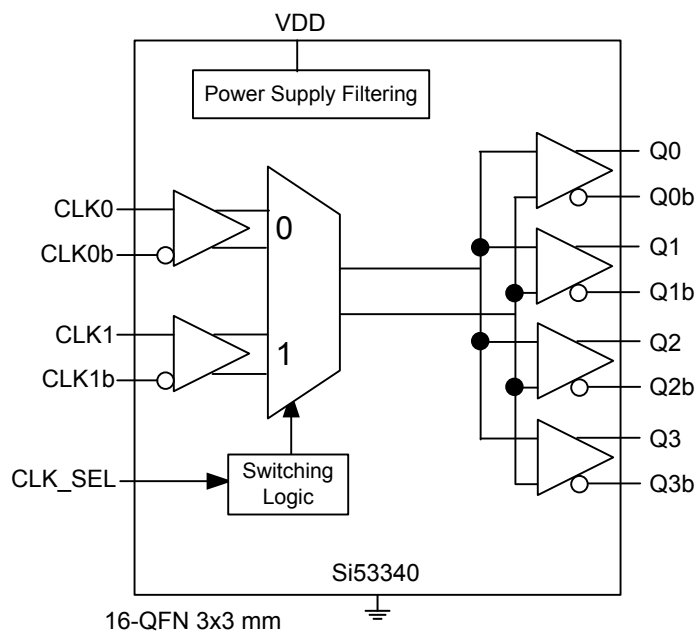


Figure 4.1. Si53340 Block Diagram

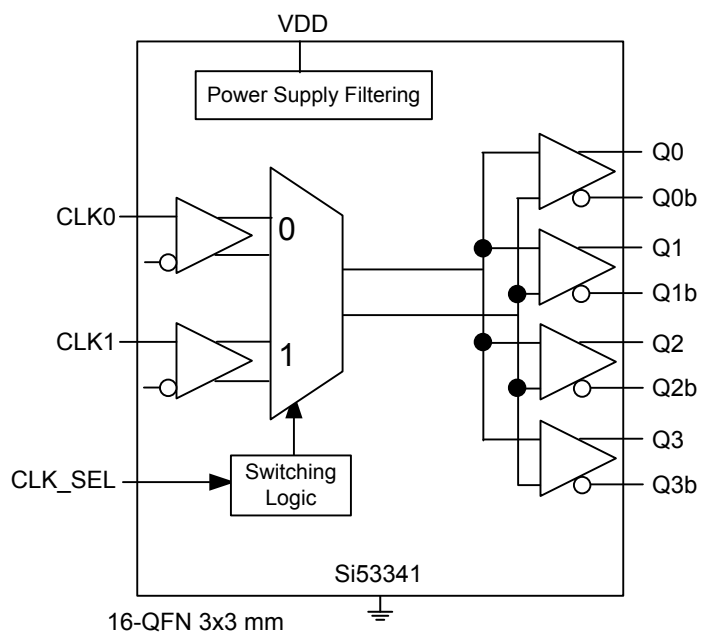


Figure 4.2. Si53341 Block Diagram

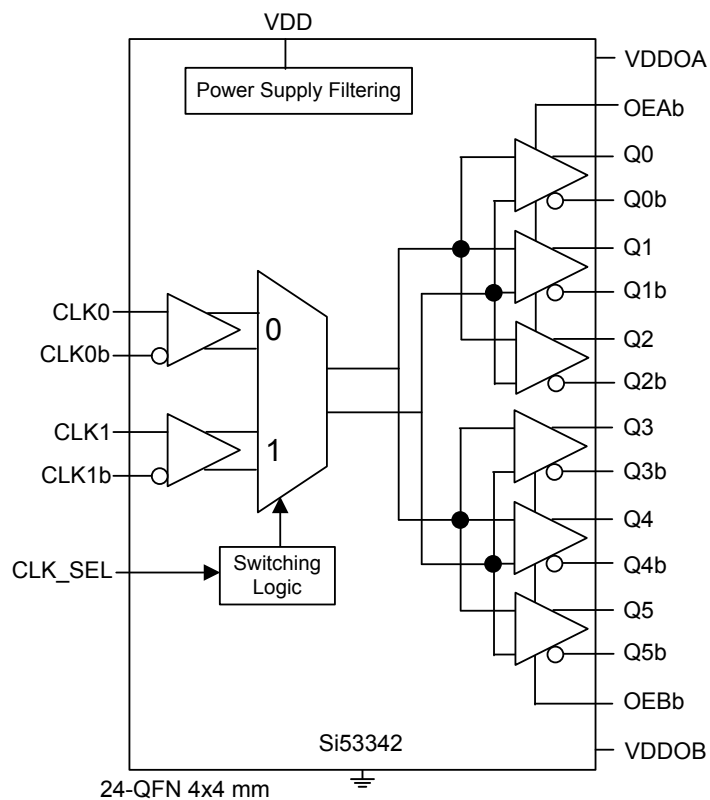


Figure 4.3. Si53342 Block Diagram

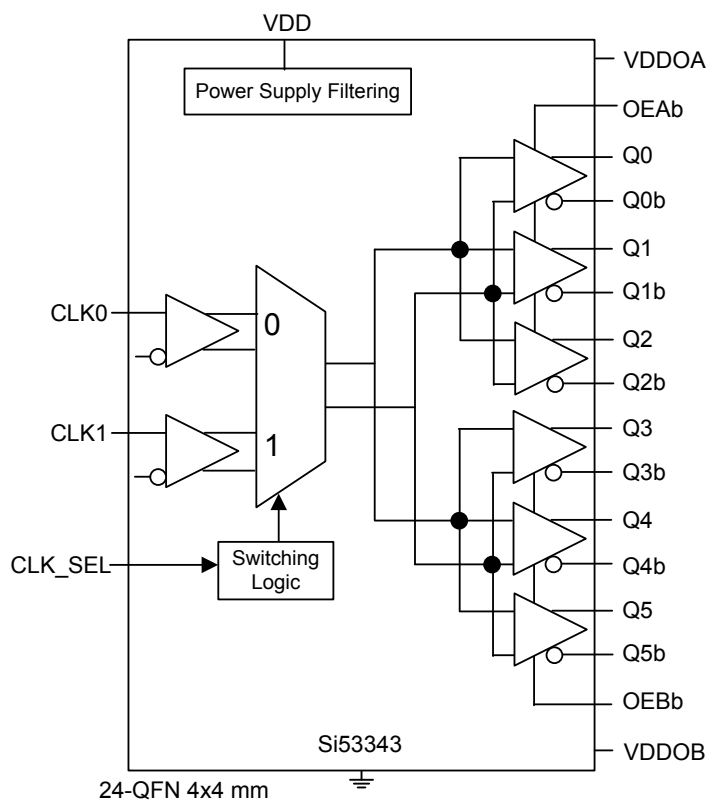


Figure 4.4. Si53343 Block Diagram

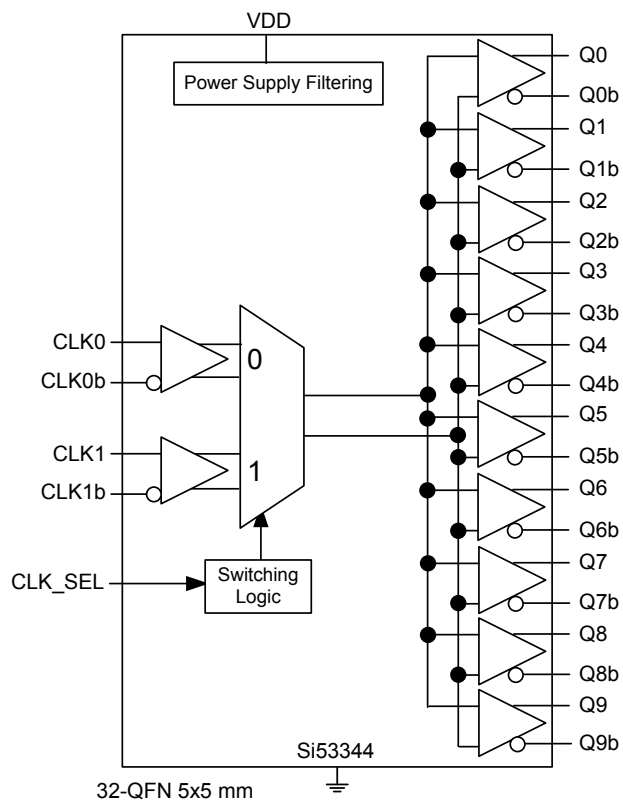


Figure 4.5. Si53344 Block Diagram

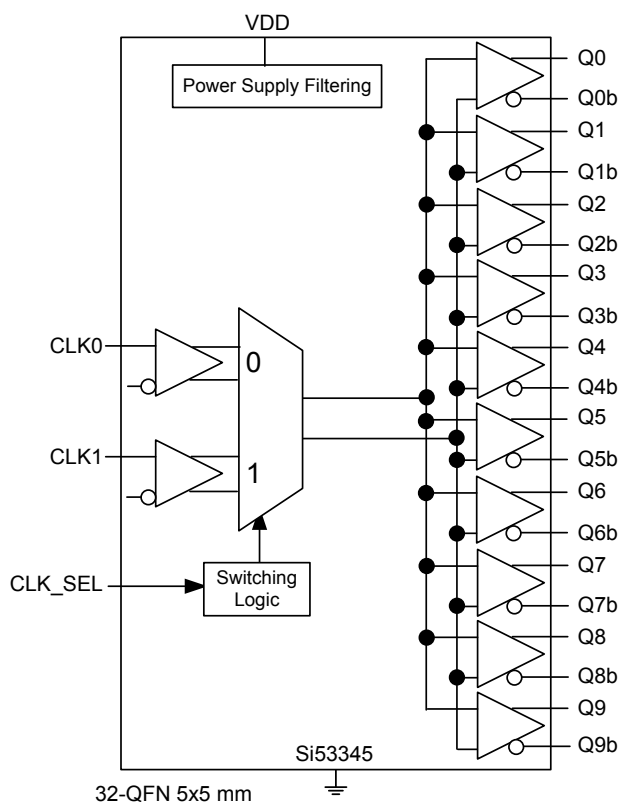


Figure 4.6. Si53345 Block Diagram

5. Pin Descriptions

5.1 Si53340/41 Pin Descriptions

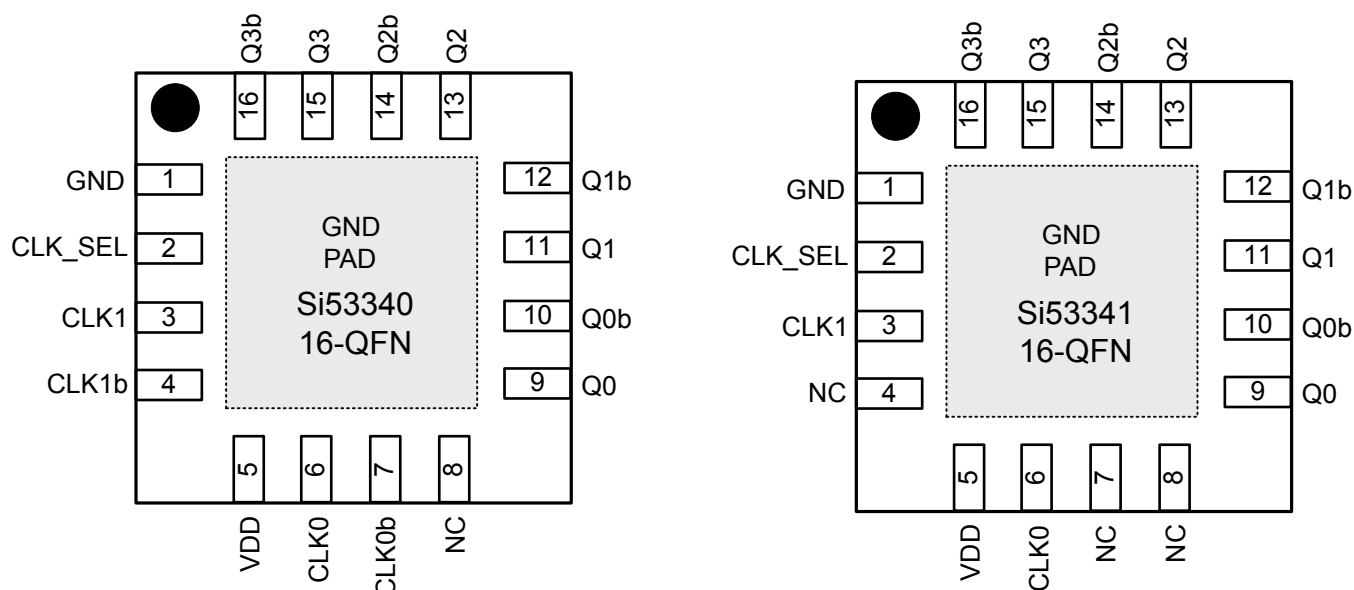


Table 5.1. Si53340/41 16-QFN Pin Descriptions

Pin	Name	Type ¹	Description
1	GND	GND	Ground.
2	CLK_SEL	I	Mux input select pin (LVCMOS). When CLK_SEL is high, CLK1 is selected. When CLK_SEL is low, CLK0 is selected. CLK_SEL contains an internal pull-down resistor.
3	CLK1	I	Input clock 1.
4	CLK1b (Si53340 only)	I	Input clock 1 (complement). When CLK1 is driven by a single-ended LVCMOS input, connect CLK1b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53341 only)	—	No connect. Leave this pin unconnected.
5	VDD	P	Core and Output Voltage Supply. Bypass with 1.0 μ F capacitor and place as close to the VDD pin as possible.
6	CLK0	I	Input Clock 0.
7	CLK0b (Si53340 only)	I	Input clock 0 (complement). When CLK0 is driven by a single-ended LVCMOS input, connect CLK0b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53341 only)	—	No connect. Leave this pin unconnected.
8	NC	—	No connect. Do not connect this pin.
9	Q0	O	Output clock 0.
10	Q0b	O	Output clock 0 (complement).
11	Q1	O	Output clock 1.
12	Q1b	O	Output clock 1 (complement).
13	Q2	O	Output clock 2.

Pin	Name	Type ¹	Description
14	Q2b	O	Output clock 2 (complement).
15	Q3	O	Output clock 3.
16	Q3b	O	Output clock 3 (complement).
GND Pad	Exposed Ground Pad	GND	Power supply ground and thermal relief. The exposed ground pad is thermally connected to the die to improve heat transfer from the package. The ground pad must be connected to GND to ensure device specifications are met.

Note:

1. I = Input; O = Output; P = Power; GND = Ground.

5.2 Si53342/43 Pin Descriptions

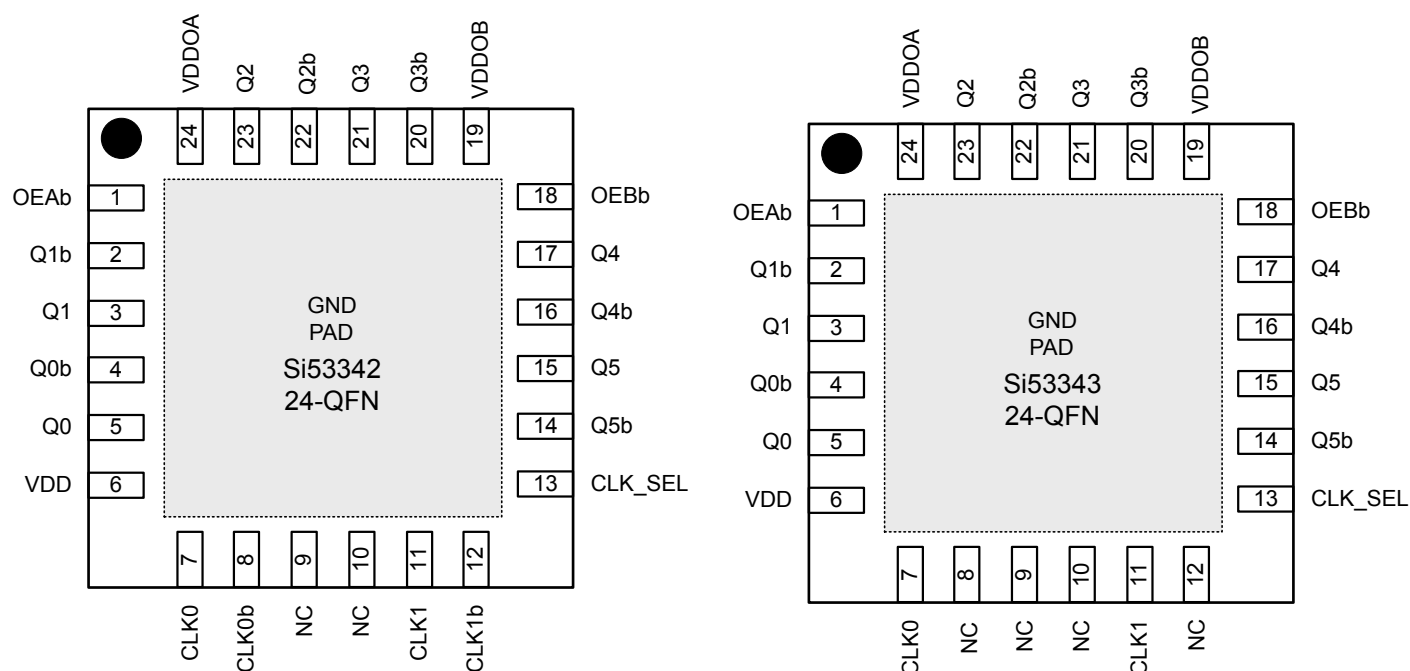


Table 5.2. Si53342/43 24-QFN Pin Descriptions

Pin	Name	Type ¹	Description
1	OEAb	I	Output Enable for Bank A (Q0, Q1, Q2). When OEAb = LOW, outputs Q0, Q1, and Q2 are enabled. This pin contains an active pull-down resistor, and leaving the pin disconnected enables the outputs. When OEAb = HIGH, Q0, Q1, and Q2 are disabled.
2	Q1b	O	Output clock 1 (complement).
3	Q1	O	Output clock 1.
4	Q0b	O	Output clock 0 (complement)
5	Q0	O	Output clock 0
6	VDD	P	Core voltage supply. Bypass with 1.0 μ F capacitor and place as close to the VDD pin as possible.
7	CLK0	I	Input clock 0.
8	CLK0b (Si53342 only)	O	Input clock 0 (complement). When CLK0 is driven by a single-ended LVCMOS input, connect CLK0b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53343 only)	—	No connect. Leave this pin unconnected.
9	NC	—	No Connect. Do not connect this pin to anything.
10	NC	—	No Connect. Do not connect this pin to anything.
11	CLK1	I	Input clock 1.
12	CLK1b (Si53342 only)	I	Input clock 1 (complement). When CLK1 is driven by a single-ended LVCMOS input, connect CLK1b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53343 only)	—	No connect. Leave this pin unconnected.

Pin	Name	Type ¹	Description
13	CLK_SEL	I	Mux input select pin (LVCMOS). When CLK_SEL is high, CLK1 is selected. When CLK_SEL is low, CLK0 is selected. CLK_SEL contains an internal pull-down resistor.
14	Q5b	O	Output clock 5 (complement).
15	Q5	O	Output clock 5.
16	Q4b	O	Output clock 4 (complement).
17	Q4	O	Output clock 4.
18	OEBb	I	Output Enable for Bank B (Q3, Q4, Q5). When OEBb = LOW, outputs Q3, Q4, and Q5 are enabled. This pin contains an active pull-down resistor, and leaving the pin disconnected enables the outputs. When OEBb = HIGH, Q3, Q4, and Q5 are disabled.
19	VDDOB	P	Output voltage supply—Bank B (Outputs: Q3 to Q5). Bypass with 1.0 μ F capacitor and place as close to the VDDOB pin as possible.
20	Q3b	O	Output clock 3 (complement).
21	Q3	O	Output clock 3.
22	Q2b	O	Output clock 2 (complement).
23	Q2	O	Output clock 2.
24	VDDOA	P	Output voltage supply—Bank A (Outputs: Q0 to Q2). Bypass with 1.0 μ F capacitor and place as close to the VDDOA pin as possible.
GND Pad	Exposed Ground Pad	GND	Power supply ground and thermal relief. The exposed ground pad is thermally connected to the die to improve heat transfer from the package. The ground pad must be connected to GND to ensure device specifications are met.

Note:

1. I = Input; O = Output; P = Power; GND = Ground.

5.3 Si53344/45 Pin Descriptions

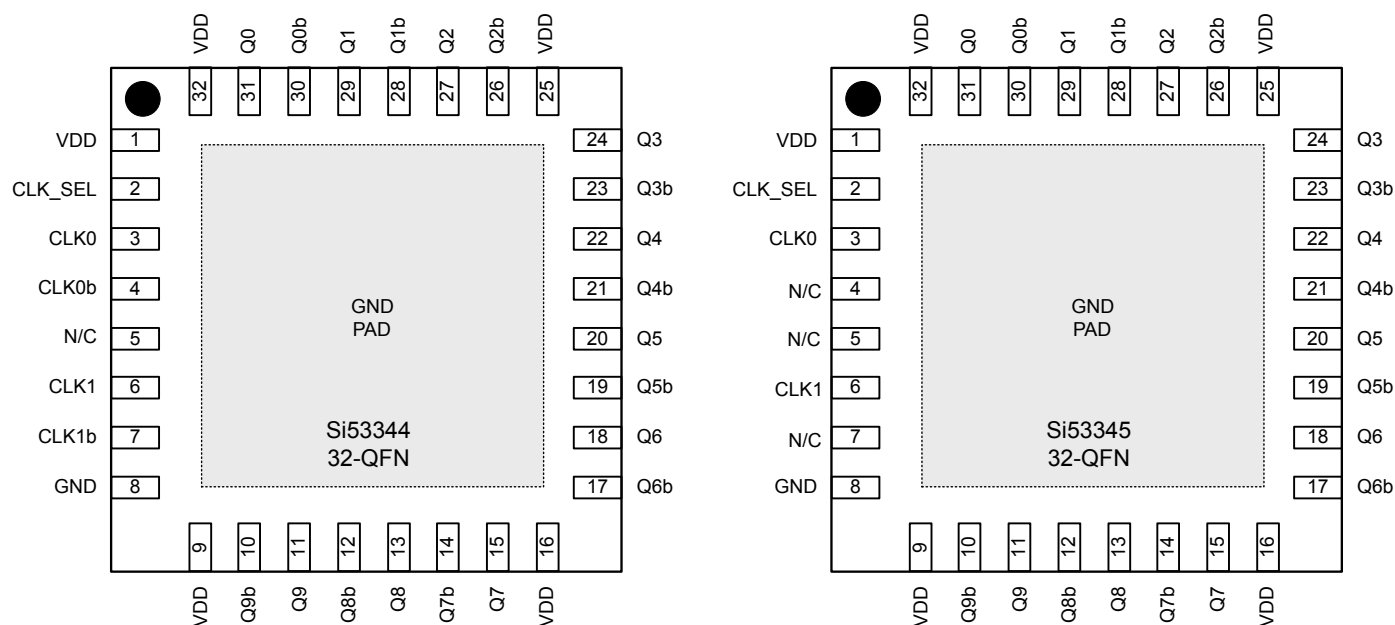


Table 5.3. Si53344/45 32-QFN Pin Descriptions

Pin #	Name	Type ¹	Description
1	VDD	P	Core and Output voltage supply. Bypass with 1.0 μ F capacitor and place as close to the VDD pin as possible.
2	CLK_SEL	I	Mux input select pin (LVCMOS). When CLK_SEL is high, CLK1 is selected. When CLK_SEL is low, CLK0 is selected. CLK_SEL contains an internal pull-down resistor.
3	CLK0	I	Input clock 0.
4	CLK0b (Si53344 only)	I	Input clock 0 (complement). When CLK0 is driven by a single-ended LVCMOS input, connect CLK0b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53345 only)	—	No connect. Leave this pin unconnected.
5	NC		No connect. Leave this pin unconnected.
6	CLK1	I	Input clock 1.
7	CLK1b (Si53344 only)	I	Input clock 1 (complement). When CLK1 is driven by a single-ended LVCMOS input, connect CLK1b to an appropriate bias voltage (e.g., $V_{DD}/2$).
	NC (Si53345 only)	—	No connect. Leave this pin unconnected.
8	GND	GND	Ground.
9	VDD	P	Core and Output voltage supply. Bypass with 1.0 μ F capacitor and place as closely to the VDD pin as possible.
10	Q9b	O	Output clock 9 (complement).
11	Q9	O	Output clock 9.
12	Q8b	O	Output clock 8 (complement).
13	Q8	O	Output clock 8.
14	Q7b	O	Output clock 7 (complement).

Pin #	Name	Type ¹	Description
15	Q7	O	Output clock 7.
16	VDD	P	Core and Output voltage supply. Bypass with 1.0 μ F capacitor and place as closely to the VDD pin as possible.
17	Q6b	O	Output clock 6 (complement).
18	Q6	O	Output clock 6.
19	Q5b	O	Output clock 5 (complement).
20	Q5	O	Output clock 5.
21	Q4b	O	Output clock 4 (complement).
22	Q4	O	Output clock 4.
23	Q3b	O	Output clock 3 (complement).
24	Q3	O	Output clock 3.
25	VDD	P	Core and Output voltage supply. Bypass with 1.0 μ F capacitor and place as closely to the VDD pin as possible.
26	Q2b	O	Output clock 2 (complement).
27	Q2	O	Output clock 2.
28	Q1b	O	Output clock 1 (complement).
29	Q1	O	Output clock 1.
30	Q0b	O	Output clock 0 (complement).
31	Q0	O	Output clock 0.
32	VDD	P	Core voltage supply. Bypass with 1.0 μ F capacitor and place as closely to the VDD pin as possible.
GND Pad	Exposed Ground Pad	GND	Power supply ground and thermal relief. The exposed ground pad is thermally connected to the die to improve the heat transfer out of the package. The ground pad must be connected to GND to ensure device specifications are met.

Note:

1. I = Input; O = Output; P = Power; GND = Ground.

6. Package Outlines

6.1 16-Pin QFN Package

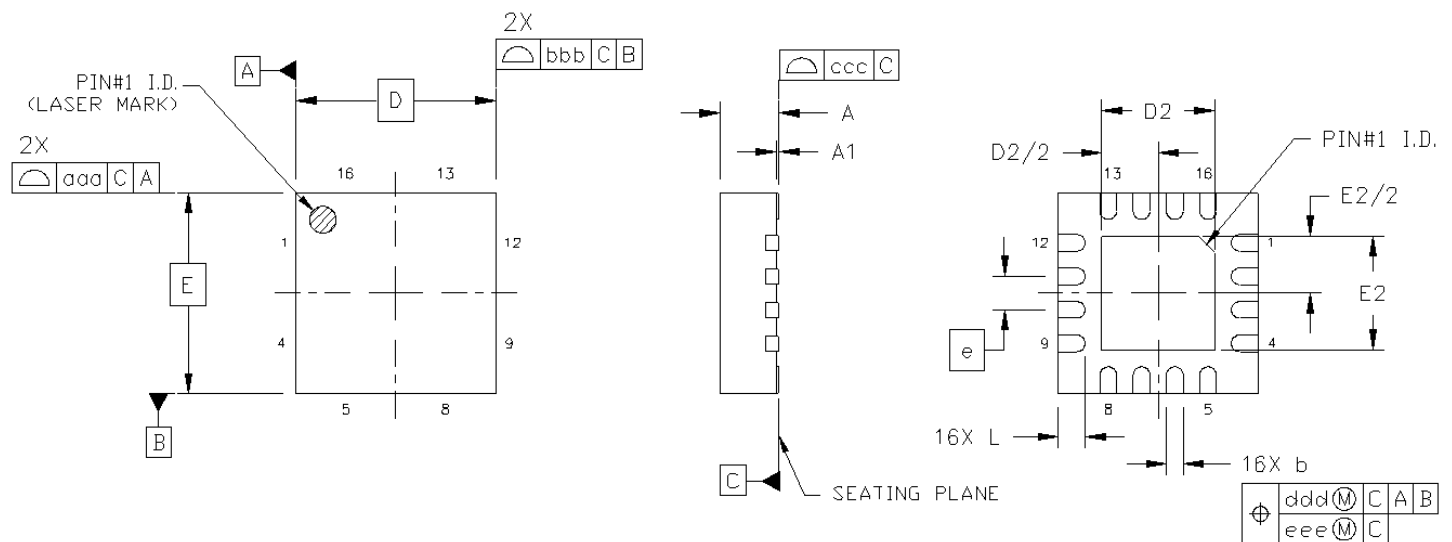


Figure 6.1. 16-Pin QFN Package

Table 6.1. 16-QFN Package Dimensions

Dimension	Min	Nom	Max
A	0.80	0.85	0.90
A1	0.00	0.02	0.05
b	0.18	0.25	0.30
D	3.00 BSC.		
D2	1.65	1.70	1.75
e	0.50 BSC.		
E	3.00 BSC.		
E2	1.65	1.70	1.75
L	0.30	0.40	0.50
aaa	—	—	0.10
bbb	—	—	0.10
ccc	—	—	0.08
ddd	—	—	0.10
eee	—	—	0.05

Note:

1. All dimensions shown are in millimeters (mm) unless otherwise noted.
2. Dimensioning and Tolerancing per ANSI Y14.5M-1994.

6.2 24-Pin QFN Package

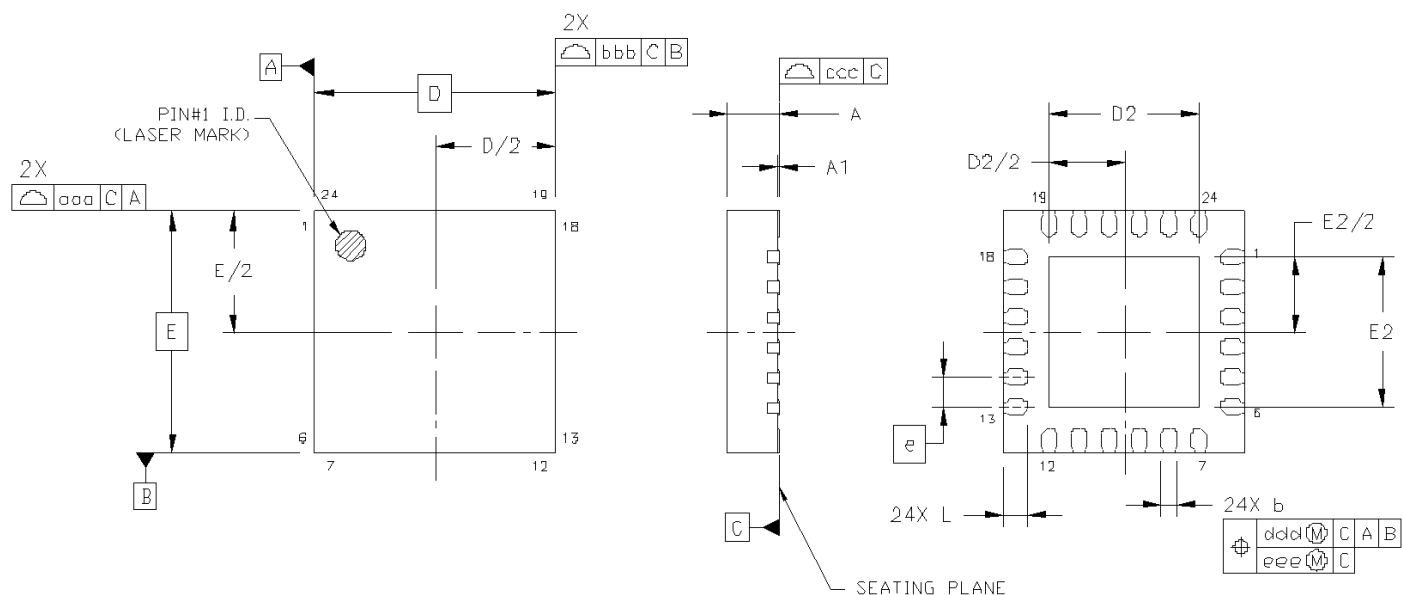


Figure 6.2. 24-Pin QFN Package

Table 6.2. 24-QFN Package Dimensions

Dimension	Min	Nom	Max
A	0.80	0.85	0.90
A1	0.00	0.02	0.05
b	0.18	0.25	0.30
D	4.00 BSC.		
D2	2.35	2.50	2.65
e	0.50 BSC.		
E	4.00 BSC.		
E2	2.35	2.50	2.65
L	0.30	0.40	0.50
aaa	0.10		
bbb	0.10		
ccc	0.08		
ddd	0.10		
eee	0.05		

Note:

1. All dimensions shown are in millimeters (mm) unless otherwise noted.
2. Dimensioning and Tolerancing per ANSI Y14.5M-1994.
3. This drawing conforms to JEDEC outline MO-220, variation VGGD-8.
4. Recommended card reflow profile is per the JEDEC/IPC J-STD-020C specification for Small Body Components.

6.3 32-Pin QFN Package

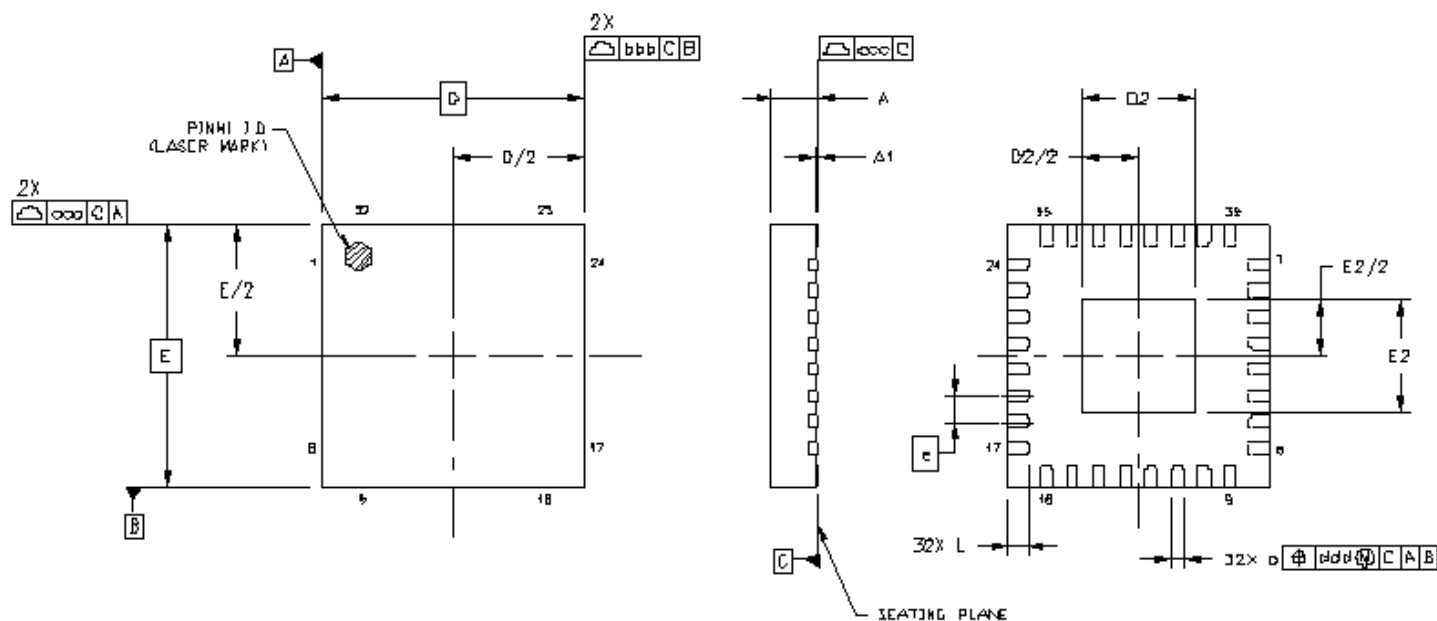


Figure 6.3. 32-Pin QFN Package

Table 6.3. 32-QFN Package Dimensions

Dimension	Min	Nom	Max
A	0.80	0.85	1.00
A1	0.00	0.02	0.05
b	0.18	0.25	0.30
c	0.20	0.25	0.30
D	5.00 BSC		
D2	2.00	2.15	2.30
e	0.50 BSC		
E	5.00 BSC		
E2	2.00	2.15	2.30
L	0.30	0.40	0.50
aaa	0.10		
bbb	0.10		
ccc	0.08		
ddd	0.10		

Note:

1. All dimensions shown are in millimeters (mm) unless otherwise noted.
2. Dimensioning and Tolerancing per ANSI Y14.5M-1994.
3. This drawing conforms to the JEDEC Solid State Outline MO-220.

7. Land Patterns

7.1 16-Pin QFN Land Pattern

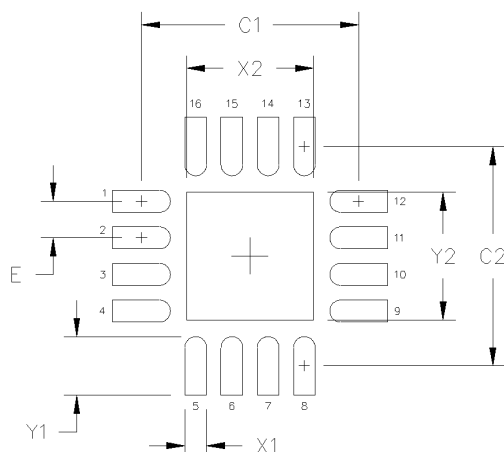


Figure 7.1. 16-Pin QFN Land Pattern

Table 7.1. 16-QFN Land Pattern Dimensions

Dimension	mm
C1	3.00
C2	3.00
E	0.50
X1	0.30
Y1	0.80
X2	1.75
Y2	1.75

Notes:

General

1. All dimensions shown are in millimeters (mm).
2. This Land Pattern Design is based on the IPC-7351 guidelines.
3. All dimensions shown are at Maximum Material Condition (MMC). Least Material Condition (LMC) is calculated based on a Fabrication Allowance of 0.05 mm.

Solder Mask Design

1. All metal pads are to be non-solder mask defined (NSMD). Clearance between the solder mask and the metal pad is to be 60 μ m minimum, all the way around the pad.

Stencil Design

1. A stainless steel, laser-cut and electro-polished stencil with trapezoidal walls should be used to assure good solder paste release.
2. The stencil thickness should be 0.125 mm (5 mils).
3. The ratio of stencil aperture to land pad size should be 1:1 for all perimeter pads.
4. A 2 x 2 array of 0.65 mm square openings on a 0.90 mm pitch should be used for the center ground pad.

Card Assembly

1. A No-Clean, Type-3 solder paste is recommended.
2. The recommended card reflow profile is per the JEDEC/IPC J-STD-020 specification for Small Body Components.

7.2 24-Pin QFN Land Pattern

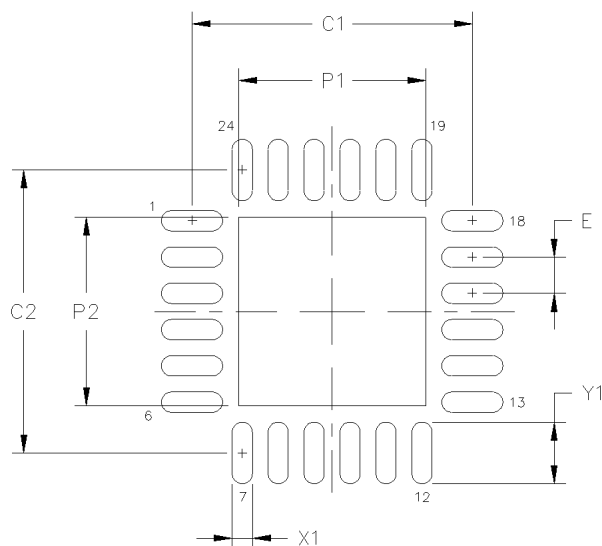


Figure 7.2. 24-Pin QFN Land Pattern

Table 7.2. 24-QFN Land Pattern Dimensions

Dimension	mm
P1	2.55
P2	2.55
X1	0.25
Y1	0.80
C1	3.90
C2	3.90
E	0.50

Notes:

General

1. All dimensions shown are in millimeters (mm).
2. This Land Pattern Design is based on the IPC-7351 guidelines.

Solder Mask Design

1. All metal pads are to be non-solder mask defined (NSMD). Clearance between the solder mask and the metal pad is to be 60 μ m minimum, all the way around the pad.

Stencil Design

1. A stainless steel, laser-cut and electro-polished stencil with trapezoidal walls should be used to assure good solder paste release.
2. The stencil thickness should be 0.125 mm (5 mils).
3. The ratio of stencil aperture to land pad size should be 1:1 for all perimeter pads.
4. A 2 x 2 array of 1.10 mm x 1.10 mm openings on 1.30 mm pitch should be used for the center ground pad.

Card Assembly

1. A No-Clean, Type-3 solder paste is recommended.
2. The recommended card reflow profile is per the JEDEC/IPC J-STD-020 specification for Small Body Components.

7.3 32-Pin QFN Land Pattern

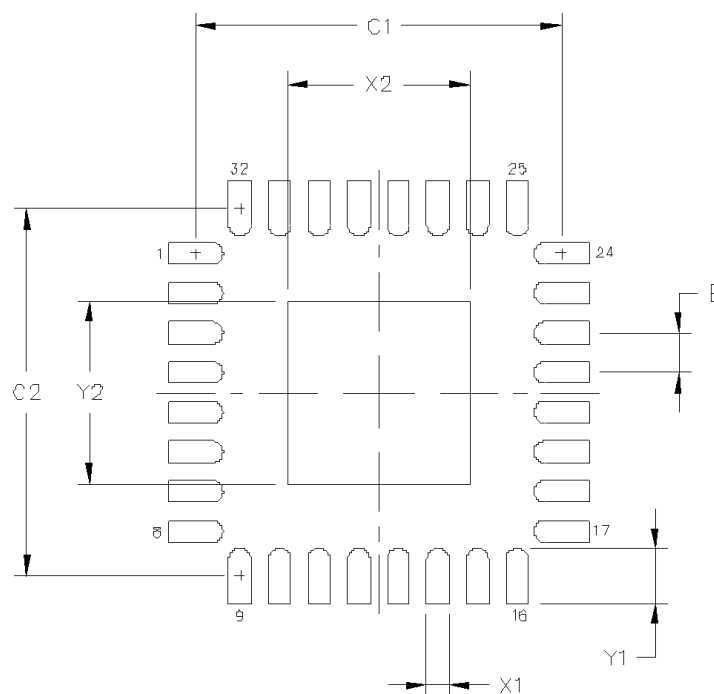


Figure 7.3. 32-Pin QFN Land Pattern

Table 7.3. 32-QFN Land Pattern Dimensions

Dimension	Min	Max		Dimension	Min	Max
C1	4.52	4.62		X2	2.20	2.30
C2	4.52	4.62		Y1	0.59	0.69
E	0.50 BSC			Y2	2.20	2.30
X1	0.20	0.30				

Notes:

General

1. All dimensions shown are in millimeters (mm) unless otherwise noted.
2. This Land Pattern Design is based on the IPC-7351 guidelines.

Solder Mask Design

1. All metal pads are to be non-solder mask defined (NSMD). Clearance between the solder mask and the metal pad is to be 60 mm minimum, all the way around the pad.

Stencil Design

1. A stainless steel, laser-cut and electro-polished stencil with trapezoidal walls should be used to assure good solder paste release.
2. The stencil thickness should be 0.125 mm (5 mils).
3. The ratio of stencil aperture to land pad size should be 1:1 for all perimeter pads.
4. A 2 x 2 array of 0.75 mm square openings on 1.15 mm pitch should be used for the center ground pad.

Card Assembly

1. A No-Clean, Type-3 solder paste is recommended.
2. The recommended card reflow profile is per the JEDEC/IPC J-STD-020 specification for Small Body Components.

8. Top Markings

8.1 Si53340/41 Top Markings

Figure 8.1. Si53340 Top Marking

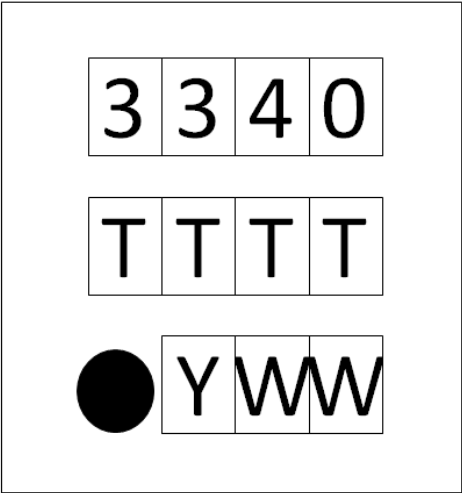


Figure 8.2. Si53341 Top Marking

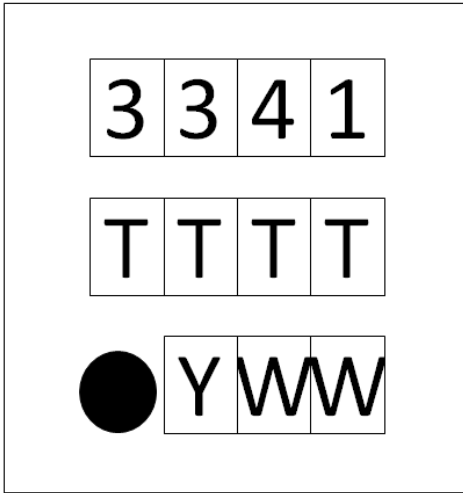


Table 8.1. Si53340/41 Top Marking Explanation

Mark Method:	Laser	
Font Size:	0.635 mm (25 mils) Right-Justified	
Line 1 Marking:	Product ID	3340 for Si53340; 3341 for Si53341
Line 2 Marking:	TTTT = Mfg Code	Manufacturing Code
Line 3 Marking	Circle = 0.5 mm Diameter Bottom-Left Justified	Pin 1 Identifier
	YWW = Date Code	Corresponds to the last digit of the current year (Y) and the work-week (WW) of the mold date.

8.2 Si53342/43 Top Markings

Figure 8.3. Si53342 Top Marking

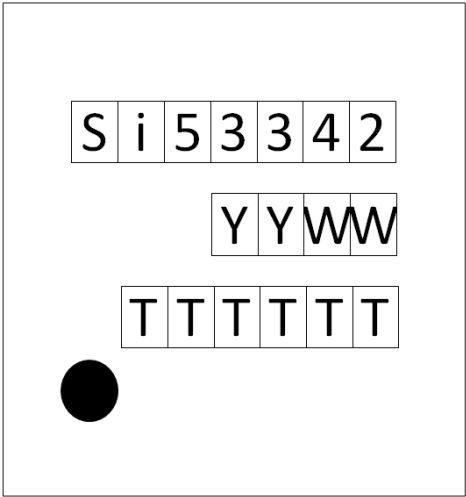


Figure 8.4. Si53343 Top Marking

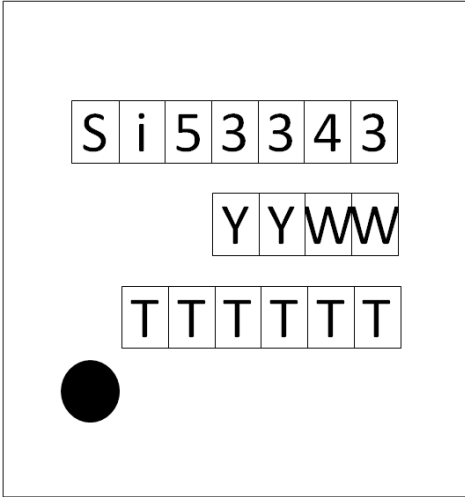


Table 8.2. Si53342/43 Top Marking Explanation

Mark Method:	Laser	
Font Size:	2.0 Point (28 mils) Center-Justified	
Line 1 Marking:	Device Part Number	Si53342 for Si53342; Si53343 for Si53343
Line 2 Marking:	Device Revision/Type	B-GM
Line 3 Marking:	TTTTTT = Mfg Code	Manufacturing Code from the Assembly Purchase Order form.
Line 4 Marking:	Circle = 0.5 mm Diameter Lower-Left Justified	Pin 1 Identifier
	YY = year WW = Work Week	Assigned by the Assembly House. Corresponds to the year and work week of the mold date.

8.3 Si53344/45 Top Markings

Figure 8.5. Si53344 Top Marking

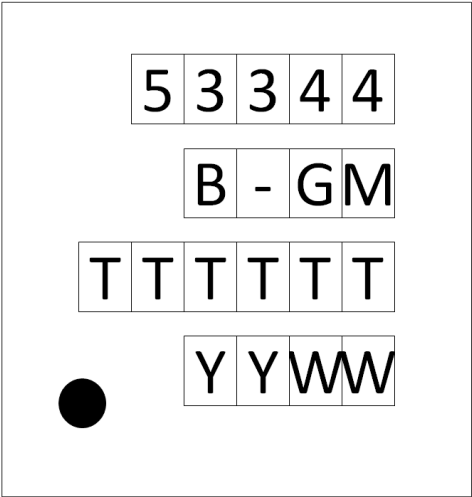


Figure 8.6. Si53345 Top Marking

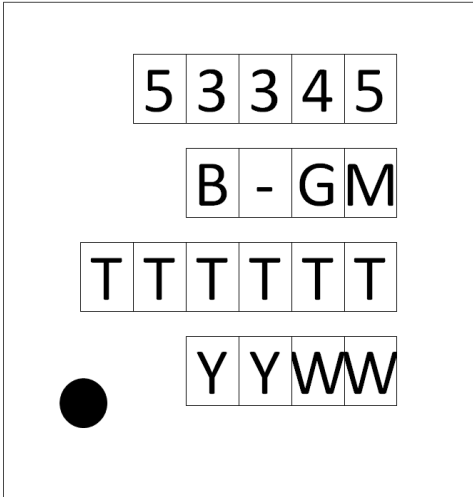


Table 8.3. Si53344/45 Top Marking Explanation

Mark Method:	Laser	
Font Size:	2.0 Point (28 mils) Center-Justified	
Line 1 Marking:	Device Part Number	53344 for Si53344; 53345 for Si53345
Line 2 Marking:	Device Revision/Type	B-GM
Line 3 Marking:	TTTTTT = Mfg Code	Manufacturing Code from the Assembly Purchase Order form.
Line 4 Marking:	Circle = 0.5 mm Diameter Lower-Left Justified	Pin 1 Identifier
	YY = year WW = Work Week	Assigned by the Assembly House. Corresponds to the year and work week of the mold date.

Table of Contents

1. Ordering Guide	2
2. Functional Description	3
2.1 Universal, Any-Format Input Termination (Si53340/42/44)	3
2.2 LVCMOS Input Termination (Si53341/43/45)	6
2.3 Input Bias Resistors	7
2.4 Input Mux.	7
2.5 Output Clock Termination Options	8
2.6 AC Timing Waveforms	9
2.7 Typical Phase Noise Performance: Differential Input Clock	10
2.8 Typical Phase Noise Performance: Single-Ended Input Clock	12
2.9 Input Mux Noise Isolation	13
2.10 Power Supply Noise Rejection	14
3. Electrical Specifications	15
4. Detailed Block Diagrams	19
5. Pin Descriptions	22
5.1 Si53340/41 Pin Descriptions	22
5.2 Si53342/43 Pin Descriptions	24
5.3 Si53344/45 Pin Descriptions	26
6. Package Outlines	28
6.1 16-Pin QFN Package	28
6.2 24-Pin QFN Package	29
6.3 32-Pin QFN Package	30
7. Land Patterns	31
7.1 16-Pin QFN Land Pattern	31
7.2 24-Pin QFN Land Pattern	32
7.3 32-Pin QFN Land Pattern	33
8. Top Markings	34
8.1 Si53340/41 Top Markings	34
8.2 Si53342/43 Top Markings	35
8.3 Si53344/45 Top Markings	36



ClockBuilder Pro

Customize Skyworks clock generators, jitter attenuators and network synchronizers with a single tool. With CBPro you can control evaluation boards, access documentation, request a custom part number, export for in-system programming and more!

www.skyworksinc.com/CBPro



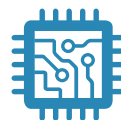
Portfolio

www.skyworksinc.com/ia/timing



SW/HW

www.skyworksinc.com/CBPro



Quality

www.skyworksinc.com/quality



Support & Resources

www.skyworksinc.com/support

Copyright © 2021 Skyworks Solutions, Inc. All Rights Reserved.

Information in this document is provided in connection with Skyworks Solutions, Inc. ("Skyworks") products or services. These materials, including the information contained herein, are provided by Skyworks as a service to its customers and may be used for informational purposes only by the customer. Skyworks assumes no responsibility for errors or omissions in these materials or the information contained herein. Skyworks may change its documentation, products, services, specifications or product descriptions at any time, without notice. Skyworks makes no commitment to update the materials or information and shall have no responsibility whatsoever for conflicts, incompatibilities, or other difficulties arising from any future changes.

No license, whether express, implied, by estoppel or otherwise, is granted to any intellectual property rights by this document. Skyworks assumes no liability for any materials, products or information provided hereunder, including the sale, distribution, reproduction or use of Skyworks products, information or materials, except as may be provided in Skyworks' Terms and Conditions of Sale.

THE MATERIALS, PRODUCTS AND INFORMATION ARE PROVIDED "AS IS" WITHOUT WARRANTY OF ANY KIND, WHETHER EXPRESS, IMPLIED, STATUTORY, OR OTHERWISE, INCLUDING FITNESS FOR A PARTICULAR PURPOSE OR USE, MERCHANTABILITY, PERFORMANCE, QUALITY OR NON-INFRINGEMENT OF ANY INTELLECTUAL PROPERTY RIGHT; ALL SUCH WARRANTIES ARE HEREBY EXPRESSLY DISCLAIMED. SKYWORKS DOES NOT WARRANT THE ACCURACY OR COMPLETENESS OF THE INFORMATION, TEXT, GRAPHICS OR OTHER ITEMS CONTAINED WITHIN THESE MATERIALS. SKYWORKS SHALL NOT BE LIABLE FOR ANY DAMAGES, INCLUDING BUT NOT LIMITED TO ANY SPECIAL, INDIRECT, INCIDENTAL, STATUTORY, OR CONSEQUENTIAL DAMAGES, INCLUDING WITHOUT LIMITATION, LOST REVENUES OR LOST PROFITS THAT MAY RESULT FROM THE USE OF THE MATERIALS OR INFORMATION, WHETHER OR NOT THE RECIPIENT OF MATERIALS HAS BEEN ADVISED OF THE POSSIBILITY OF SUCH DAMAGE.

Skyworks products are not intended for use in medical, lifesaving or life-sustaining applications, or other equipment in which the failure of the Skyworks products could lead to personal injury, death, physical or environmental damage. Skyworks customers using or selling Skyworks products for use in such applications do so at their own risk and agree to fully indemnify Skyworks for any damages resulting from such improper use or sale.

Customers are responsible for their products and applications using Skyworks products, which may deviate from published specifications as a result of design defects, errors, or operation of products outside of published parameters or design specifications. Customers should include design and operating safeguards to minimize these and other risks. Skyworks assumes no liability for applications assistance, customer product design, or damage to any equipment resulting from the use of Skyworks products outside of Skyworks' published specifications or parameters.

Skyworks, the Skyworks symbol, Sky5®, SkyOne®, SkyBlue™, Skyworks Green™, Clockbuilder®, DSPLL®, ISOModem®, ProSLIC®, and SiPHY® are trademarks or registered trademarks of Skyworks Solutions, Inc. or its subsidiaries in the United States and other countries. Third-party brands and names are for identification purposes only and are the property of their respective owners. Additional information, including relevant terms and conditions, posted at www.skyworksinc.com, are incorporated by reference.

Skyworks Solutions, Inc. | Nasdaq: SWKS | sales@skyworksinc.com | www.skyworksinc.com

USA: 781-376-3000 | Asia: 886-2-2735 0399 | Europe: 33 (0)1 43548540 | [in](#) [f](#) [t](#) [v](#)